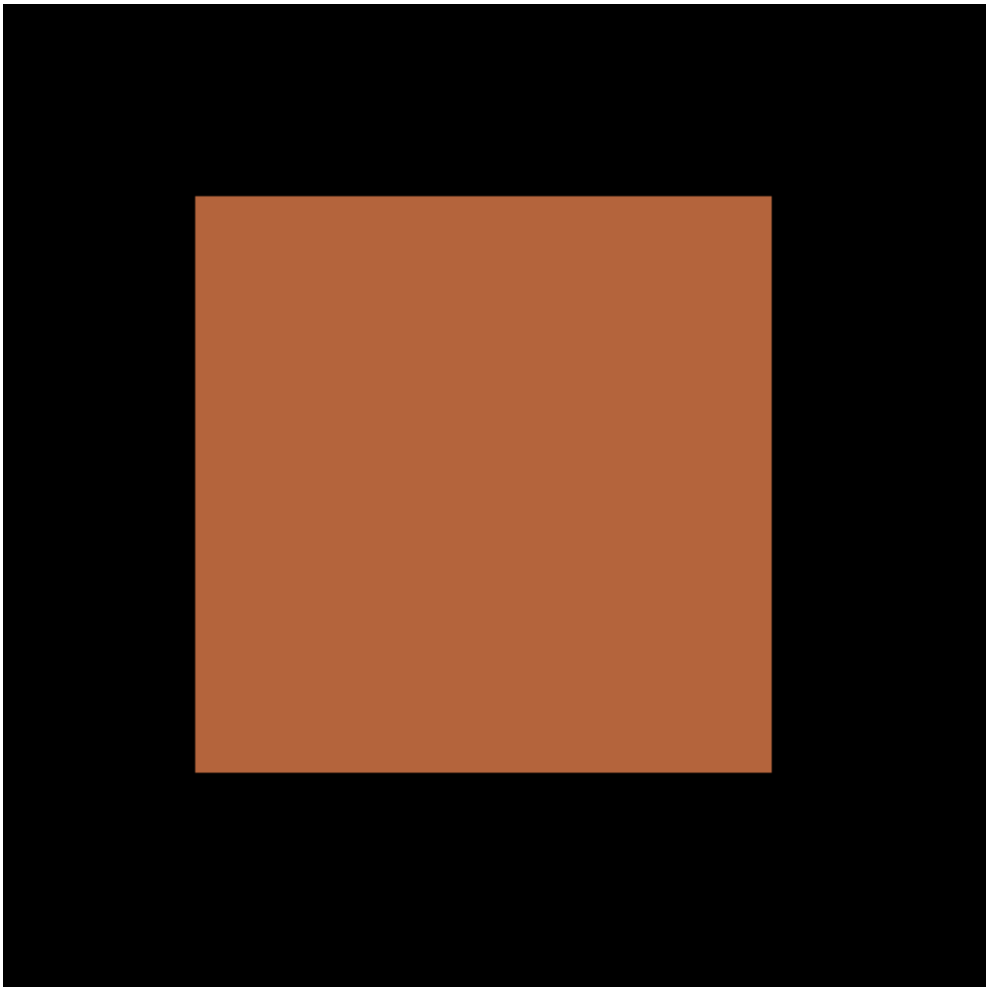


Aakhi — Retinal Image Analysis Report

Phase 1 Report (DR, OD/OC, Lesions, Glaucoma)

Patient Name	Test Patient	Doctor	Dr. Test Doctor
Age	45	Date	08 May 2026, 10:12 AM
Gender	Male	Eye Examined	Right Eye (OD)
Report ID	0241E58F	Institution	IIT Bhubaneswar — Eye AI Lab

Fundus Image



Original fundus photograph

DR Grading

Classification: **Mild DR** (Level 1/4)

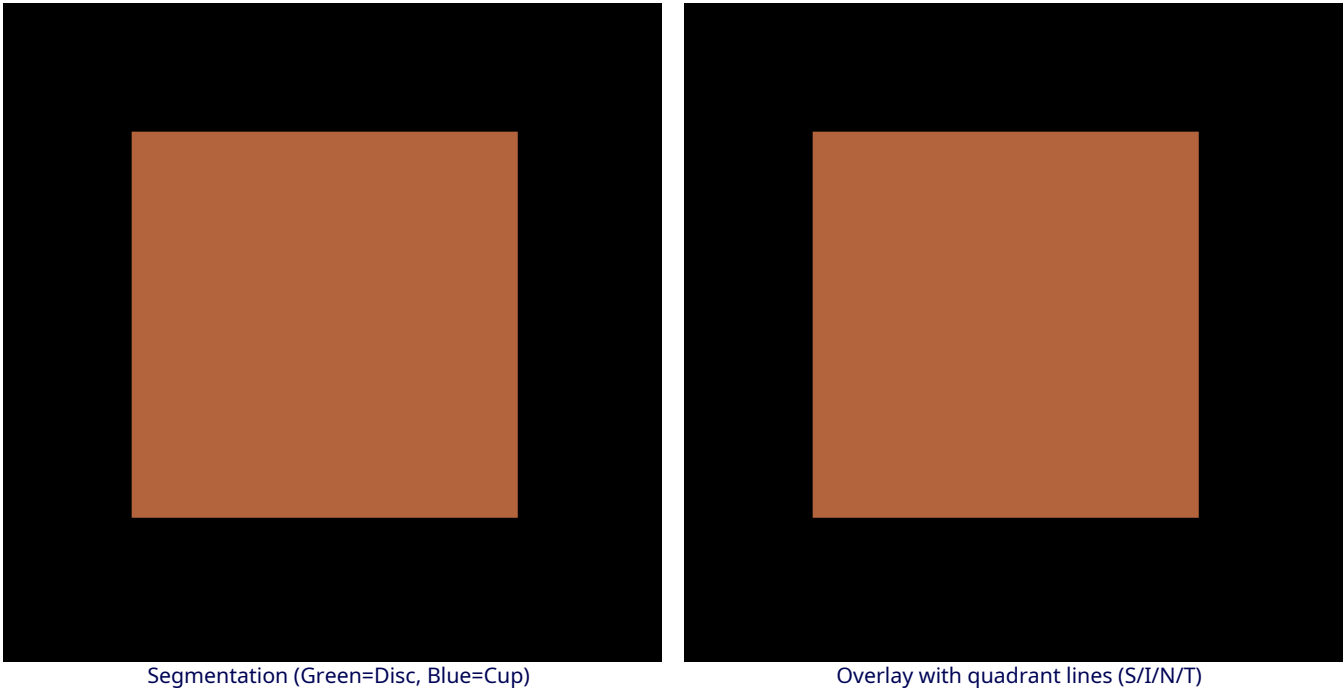
Clinical Description: Mild non-proliferative DR. Presence of at least one microaneurysm. Regular monitoring recommended.

DR Severity Level	1 / 4 — Mild DR
Grading Scale	ETDRS / International Clinical DR Disease Severity Scale
Model	EfficientNet-B6 (fine-tuned, 5-class)

Recommended Action	Follow-up in 6–12 months. Optimise glycaemic and blood pressure control.
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Optic Disc / Cup Segmentation

Color Key: Green=Optic Disc | Blue=Optic Cup | Quadrant Analysis (S=Superior, I=Inferior, N=Nasal, T=Temporal)



Morphometric Measurements

Vertical CDR (vCDR)	0.450 — CDR within normal limits (< 0.5). Low suspicion for glaucomatous optic neuropathy.
Horizontal CDR (hCDR)	0.400
Area-based CDR	0.380
Optic Disc — Vertical Ø	400 px
Optic Disc — Horizontal Ø	420 px
Optic Cup — Vertical Ø	180 px
Optic Cup — Horizontal Ø	168 px
Disc Area (px²)	125664
Cup Area (px²)	25447

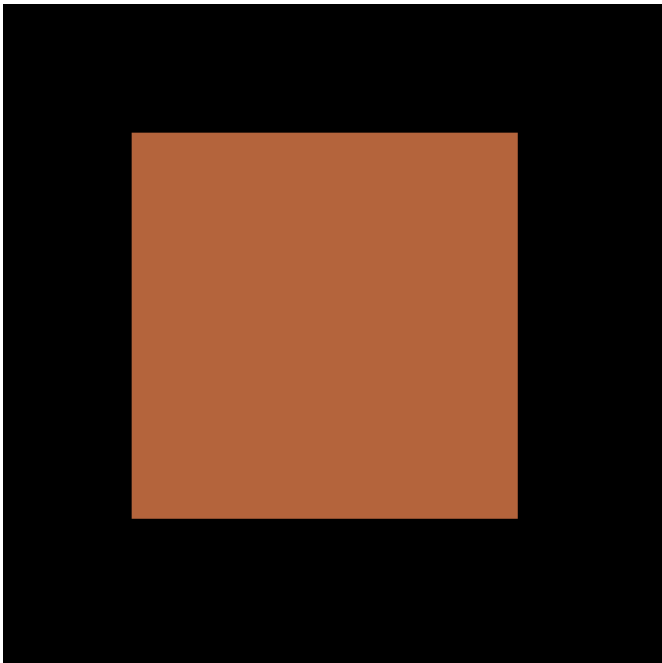
Neuroretinal Rim Analysis — ISNT Rule

Normal ISNT rule: Inferior rim Superior Nasal Temporal

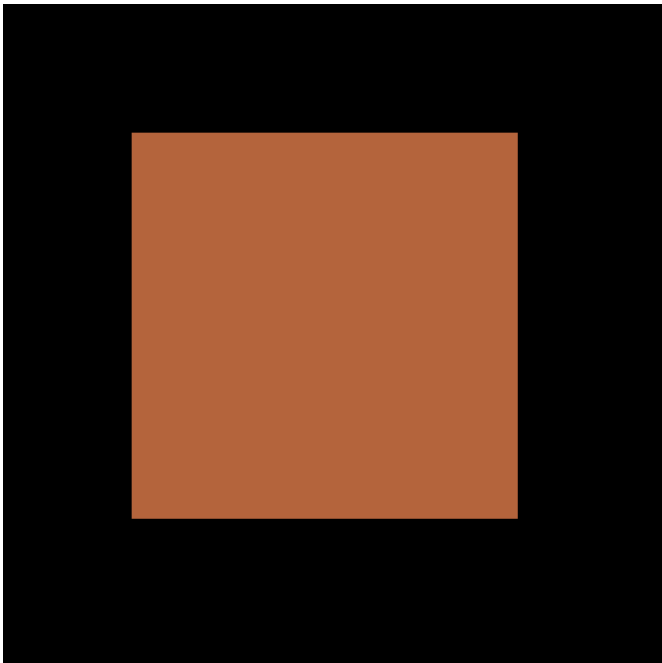
Superior Rim	28.2% of total rim area
Inferior Rim	29.1% of total rim area
Nasal Rim	21.4% of total rim area
Temporal Rim	21.3% of total rim area
ISNT Rule	Normal (ISNT rule satisfied)

Lesion Detection

Color Key: Red=Hard Exudates | Green=Hemorrhages | Blue=Microaneurysms | Yellow=Soft Exudates



Lesion segmentation (colour-coded by type)



Overlay on original fundus

Detected Lesion Areas (pixels²)

Hard Exudates (HE)	500 px ² — Lipid deposits from leaking vessels
Haemorrhages (HEM)	200 px ² — Intraretinal bleeding
Microaneurysms (MA)	100 px ² — Small capillary bulges
Soft Exudates (SE)	50 px ² — Cotton-wool spots
Model	UNet + FIAM attention (IDRiD dataset)

Glaucoma Grading

Glaucoma Severity Grade: No Glaucoma

Clinical Interpretation: No evidence of glaucomatous optic neuropathy.

Model	SE-ResNet101 (3-class glaucoma grading)
Classification	No Glaucoma
Corroborating CDR	vCDR = 0.450 (suspicion threshold: 0.65)
ISNT Rule Violation	No

Clinical Summary

DR Grading	Mild DR (Level 1/4)
Glaucoma Grading	No Glaucoma
Vertical CDR	0.450 — Risk: Low
ISNT Rule	Normal
Lesion Load	HE: 500 HEM: 200 SE: 50 px ²

Recommendations

- DR follow-up: Follow-up in 6–12 months. Optimise glycaemic and blood pressure control.

Report ID: 0241E58F | 08 May 2026, 10:12 AM

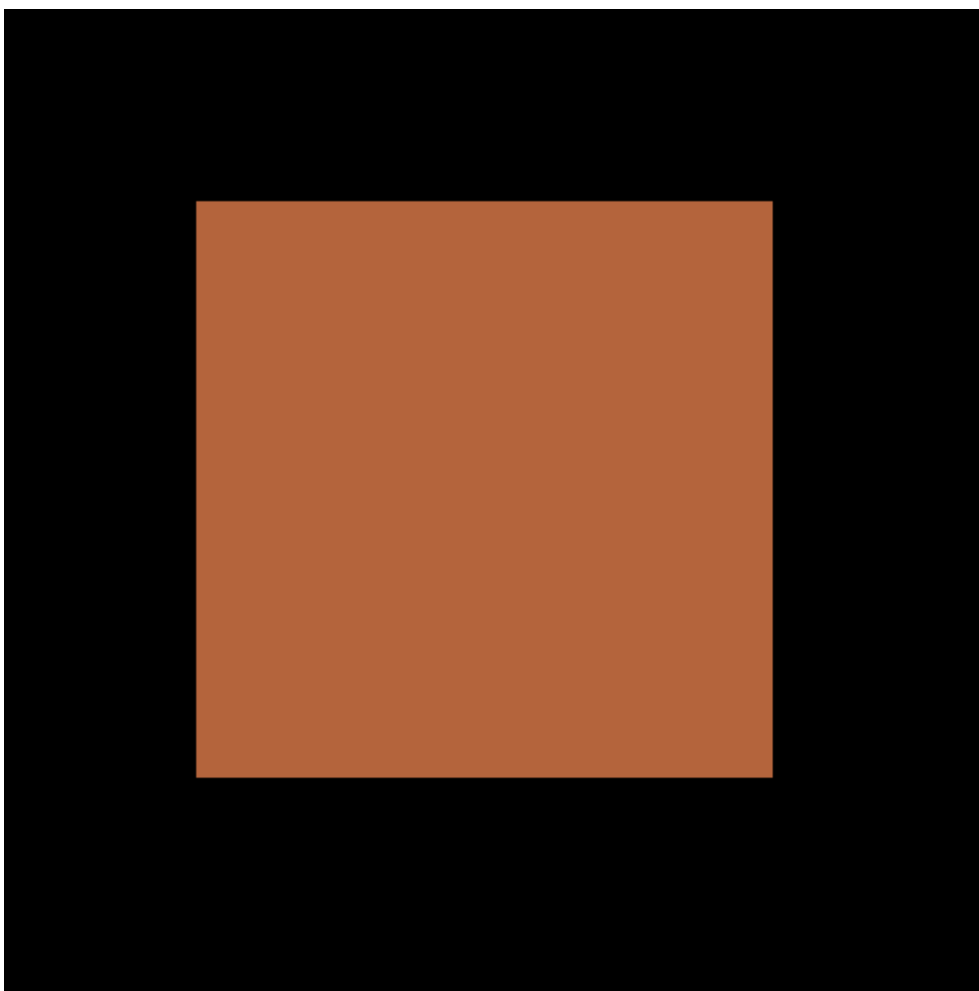
Generated by Aakhi Retinal AI

This report is generated by the Aakhi AI system for clinical reference only. All findings must be verified by a qualified ophthalmologist before treatment decisions.

आँखी — रेटनिल छव विश्लेषण रपिर्स्ट

चरण 1 रपिर्स्ट (, /, घाव, ग्लूकोमा)

रोगी का नाम	Test Patient	डॉक्टर	Dr. Test Doctor
आयु	45	दिनांक	08 May 2026, 10:12 AM
लिंग	Male	जांची गई आँख	Right Eye (OD)
रपिर्स्ट	0241E58F		IIT Bhubaneswar — Eye AI Lab



Original fundus photograph

मधुमेह रेटिनोपैथी ग्रेडिंग

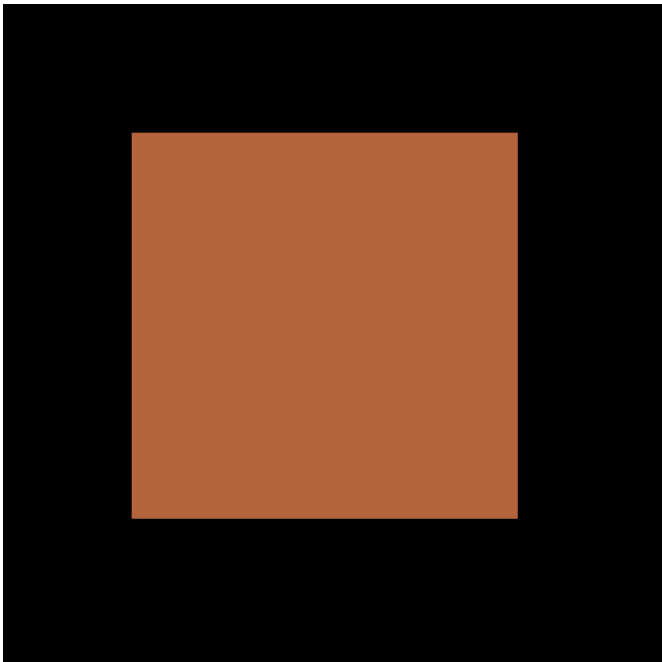
Classification: Mild DR (Level 1/4)

Clinical Description: Mild non-proliferative DR. Presence of at least one microaneurysm. Regular monitoring recommended.

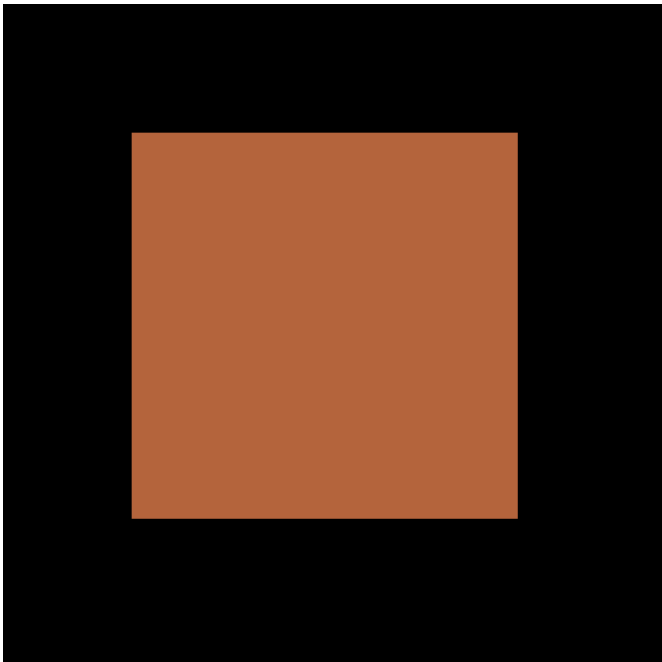
DR Severity Level	1 / 4 — Mild DR
Grading Scale	ETDRS / International Clinical DR Disease Severity Scale
Model	EfficientNet-B6 (fine-tuned, 5-class)
Recommended Action	Follow-up in 6–12 months. Optimise glycaemic and blood pressure control.

ऑप्टिक डिस्क / कप वभिजन

रंग कुंजी: हरा=ऑप्टिक डिस्क | नीला=ऑप्टिक कप | चतुर्भाग वश्लेषण (=ऊपरी, =नचिला, =नासकि, =टेम्पोरल)



Segmentation (Green=Disc, Blue=Cup)



Overlay with quadrant lines (S/I/N/T)

Morphometric Measurements

Vertical CDR (vCDR)	0.450 — CDR within normal limits (< 0.5). Low suspicion for glaucomatous optic neuropathy.
Horizontal CDR (hCDR)	0.400
Area-based CDR	0.380
Optic Disc — Vertical Ø	400 px
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Disc Area (px²)	125664
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Neuroretinal Rim Analysis — ISNT Rule

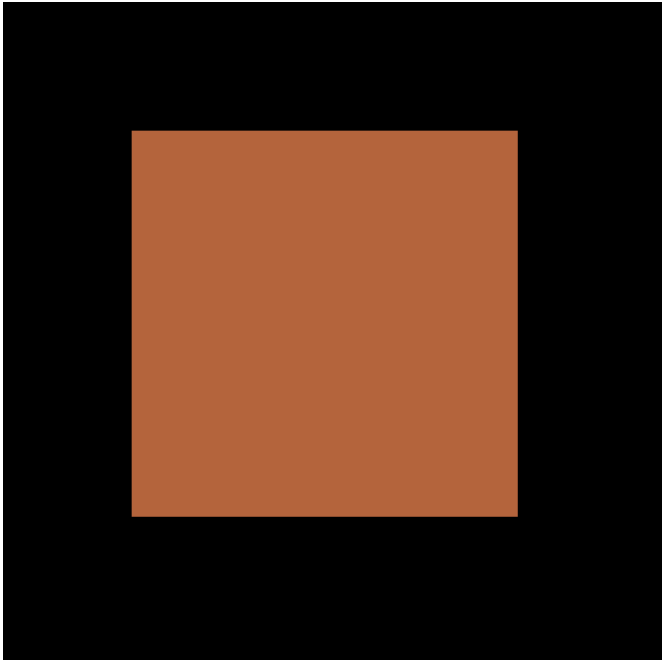
:

Superior Rim	28.2% of total rim area
Inferior Rim	29.1% of total rim area
Nasal Rim	21.4% of total rim area
Temporal Rim	21.3% of total rim area

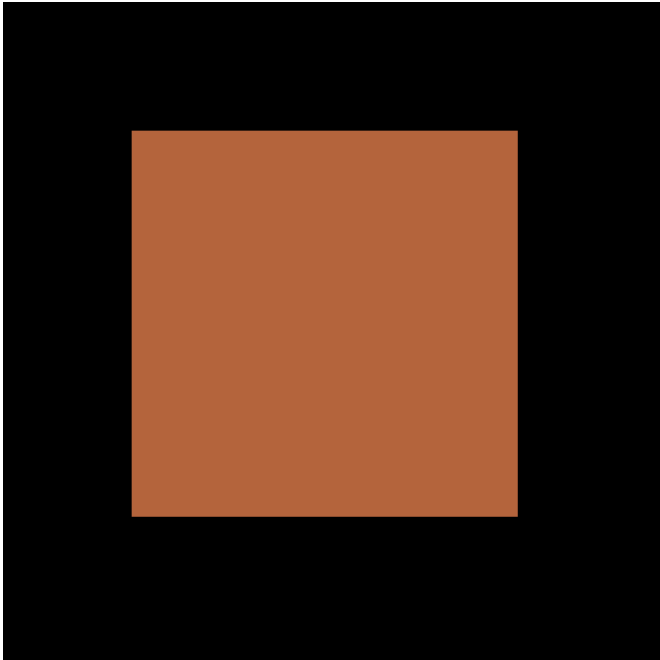
ISNT Rule	(ISNT)
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घाव पहचान

रंग कुंजी: लाल=कठोर एक्सुडेट | हरा=रक्तस्राव | नीला=माइक्रोएन्यूरिज्म | पीला=मृदु एक्सुडेट



Lesion segmentation (colour-coded by type)



Overlay on original fundus

Detected Lesion Areas (pixels²)

Hard Exudates (HE)	500 px ² — Lipid deposits from leaking vessels
Haemorrhages (HEM)	200 px ² — Intraretinal bleeding
Microaneurysms (MA)	100 px ² — Small capillary bulges
Soft Exudates (SE)	50 px ² — Cotton-wool spots
Model	UNet + FIAM attention (IDRiD dataset)

ग्लूकोमा ग्रेडिंग

Glaucoma Severity Grade: No Glaucoma

Clinical Interpretation: No evidence of glaucomatous optic neuropathy.

Model	SE-ResNet101 (3-class glaucoma grading)
Classification	No Glaucoma
Corroborating CDR	vCDR = 0.450 (suspicion threshold: 0.65)
ISNT Rule Violation	No

नैदानिक सारांश

DR Grading	Mild DR (Level 1/4)
Glaucoma Grading	No Glaucoma
Vertical CDR	0.450 — Risk: Low
ISNT Rule	Normal
Lesion Load	HE: 500 HEM: 200 SE: 50 px ²

सफारिशें

- DR follow-up: Follow-up in 6–12 months. Optimise glycaemic and blood pressure control.

Report ID: 0241E58F | 08 May 2026, 10:12 AM

आँखी रेटिनल द्वारा उत्पन्न

यह रपिर्ट केवल नैदानिक संदर्भ के लिए आँखी प्रणाली द्वारा उत्पन्न की गई है। उपचार निर्णय लेने से पहले सभी नषिकर्षों की योग्य नेत्र चिकित्सक द्वारा सत्यापन आवश्यक है।